Clinical Practice Guidelines: Management of Atrial Fibrillation

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1. Executive Summary & Clinical Scope

Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia globally, associated with a five-fold increase in the risk of stroke, a three-fold increase in heart failure risk, and elevated cardiovascular mortality. This clinical guideline delineates evidence-based diagnostic criteria, stroke risk stratification algorithms, pharmacologic rate vs. rhythm control strategies, and anticoagulation therapy recommendations.

2. Diagnostic Criteria & Initial Assessment

A definitive diagnosis of atrial fibrillation requires standard 12-lead electrocardiogram (ECG) documentation or a single-lead ECG tracing showing an irregular rhythm without discernible P waves lasting at least 30 seconds.
Initial clinical workup must include:
? 12-lead ECG assessing ventricular rate, QRS width, and ischemic changes.
? Transthoracic Echocardiogram (TTE) evaluating left ventricular ejection fraction (LVEF), left atrial dimension, and valvular heart disease.
? Laboratory panel: Thyroid stimulating hormone (TSH), complete blood count, serum creatinine, eGFR, liver transaminases, and electrolytes.

3. Thromboembolism Risk Stratification (CHA2DS2-VASc)

Oral anticoagulation decisions are guided by the validated CHA2DS2-VASc score:
? Congestive heart failure / LV dysfunction (+1 point)
? Hypertension consistently >140/90 mmHg or treated (+1 point)
? Age >= 75 years (+2 points)
? Diabetes mellitus (+1 point)
? Stroke, TIA, or prior thromboembolism (+2 points)
? Vascular disease (prior MI, peripheral artery disease, aortic plaque) (+1 point)
? Age 65 to 74 years (+1 point)
? Sex category Female (+1 point)

Clinical Recommendations:
- Score >= 2 in men or >= 3 in women: Oral anticoagulation is strongly recommended (Class I, Level A).
- Score = 1 in men or = 2 in women: Oral anticoagulation should be considered based on bleeding risk and shared decision-making.
- Score = 0 in men or = 1 in women: Anticoagulation is generally omitted (Class III: Harm).

4. Oral Anticoagulation Pharmacotherapy

Direct Oral Anticoagulants (DOACs) are first-line agents over vitamin K antagonists (Warfarin) for non-valvular AF due to superior safety profiles and significantly reduced intracranial hemorrhage risk.

Specific Dosing Guidelines:
? Apixaban (Eliquis): Standard dose is 5 mg PO twice daily (BID). Reduce dose to 2.5 mg PO BID if the patient meets at least TWO of the following ABC criteria: Age >= 80 years, Body weight <= 60 kg, or Serum creatinine >= 1.5 mg/dL.
? Rivaroxaban (Xarelto): Standard dose is 20 mg PO once daily taken with the evening meal. Dose reduced to 15 mg daily if CrCl is 15-49 mL/min.
? Dabigatran (Pradaxa): 150 mg PO twice daily. Consider 110 mg PO BID in patients >= 80 years or with elevated bleeding risk.
? Warfarin: Indicated specifically in mechanical heart valves or moderate-to-severe mitral stenosis. Target INR range is 2.0 to 3.0 (2.5 to 3.5 for mechanical mitral valves).

5. Ventricular Rate Control Strategies

For hemodynamically stable AF, ventricular rate control is initial management. Target resting heart rate is < 110 bpm (lenient control) or < 80 bpm if symptomatic.

First-line Pharmacotherapy:
? Beta-Blockers:
  - Metoprolol succinate: 50 mg to 200 mg PO once daily.
  - Bisoprolol: 2.5 mg to 10 mg PO once daily.
  - Carvedilol: 3.125 mg to 25 mg PO twice daily (preferred in heart failure with reduced ejection fraction).
? Non-Dihydropyridine Calcium Channel Blockers:
  - Diltiazem: 120 mg to 360 mg PO daily (avoid in HFrEF with LVEF < 40%).
  - Verapamil: 120 mg to 360 mg PO daily in divided doses.
? Second-line Add-on:
  - Digoxin: 0.0625 mg to 0.25 mg daily, useful in sedentary patients with concurrent HFrEF.
  - Amiodarone: Reserved for refractory rate control or when other agents are contraindicated.

6. Rhythm Control & Cardioversion Protocol

Rhythm control is indicated for patients with persistent symptoms despite rate control, younger patients, or heart failure precipitated by tachycardia.

Cardioversion Anticoagulation Protocol:
? If AF duration > 48 hours (or unknown duration): Must complete at least 3 consecutive weeks of therapeutic anticoagulation before elective cardioversion, OR undergo transesophageal echocardiography (TEE) to exclude left atrial appendage (LAA) thrombus.
? Post-cardioversion: Therapeutic anticoagulation must be maintained for a minimum of 4 weeks regardless of CHA2DS2-VASc score.